

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Dashboard Development Task

Course Code: DSA0502

Course Title: Query Processing for Data Science

CO Assessed: CO5

Assessment: TOOL 1 – Dashboard Development Task

Weightage: 25%

Total Marks:

CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication

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Reg. No.: 192324004

Section / Batch: A

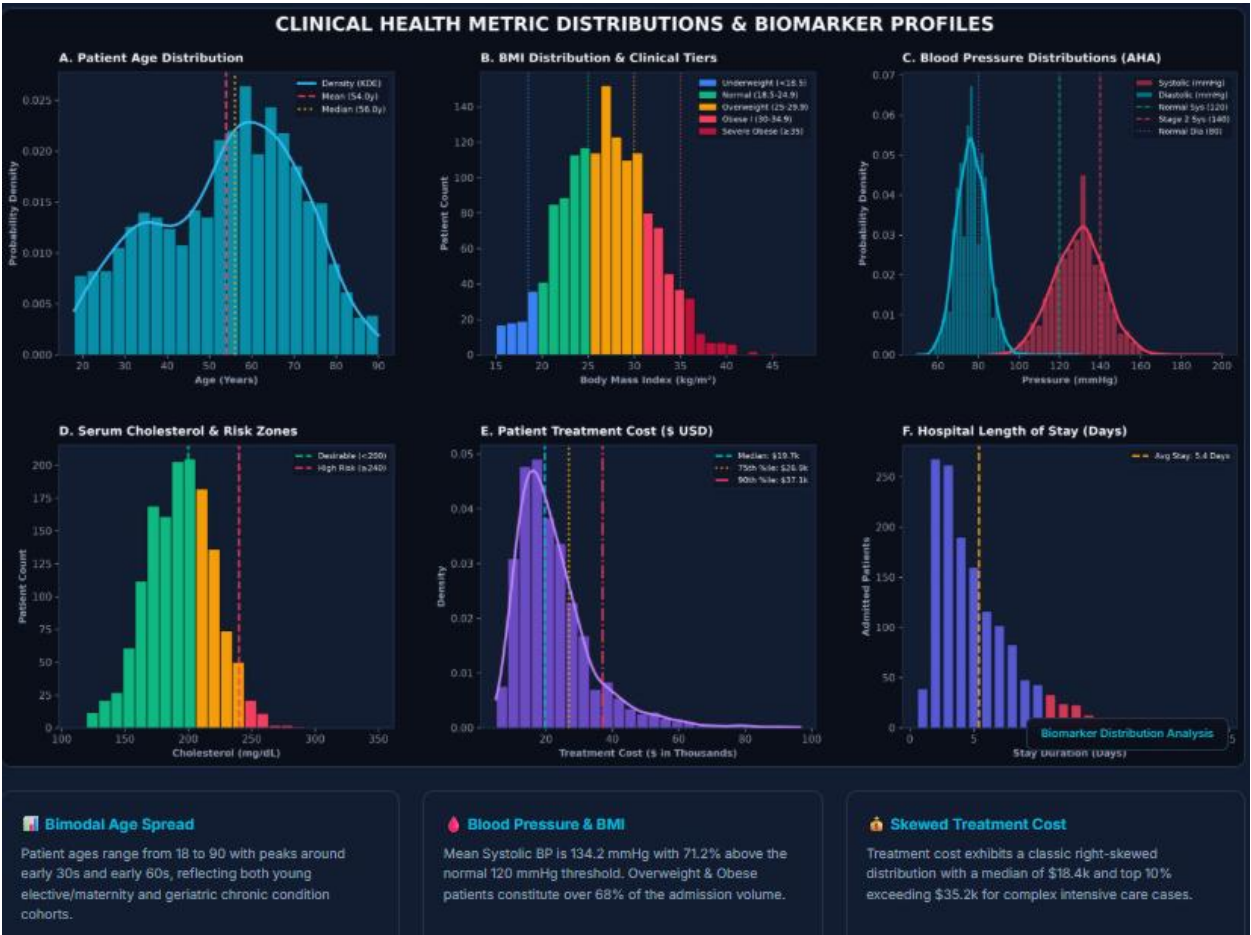
Date of Submission: 29.08.2025

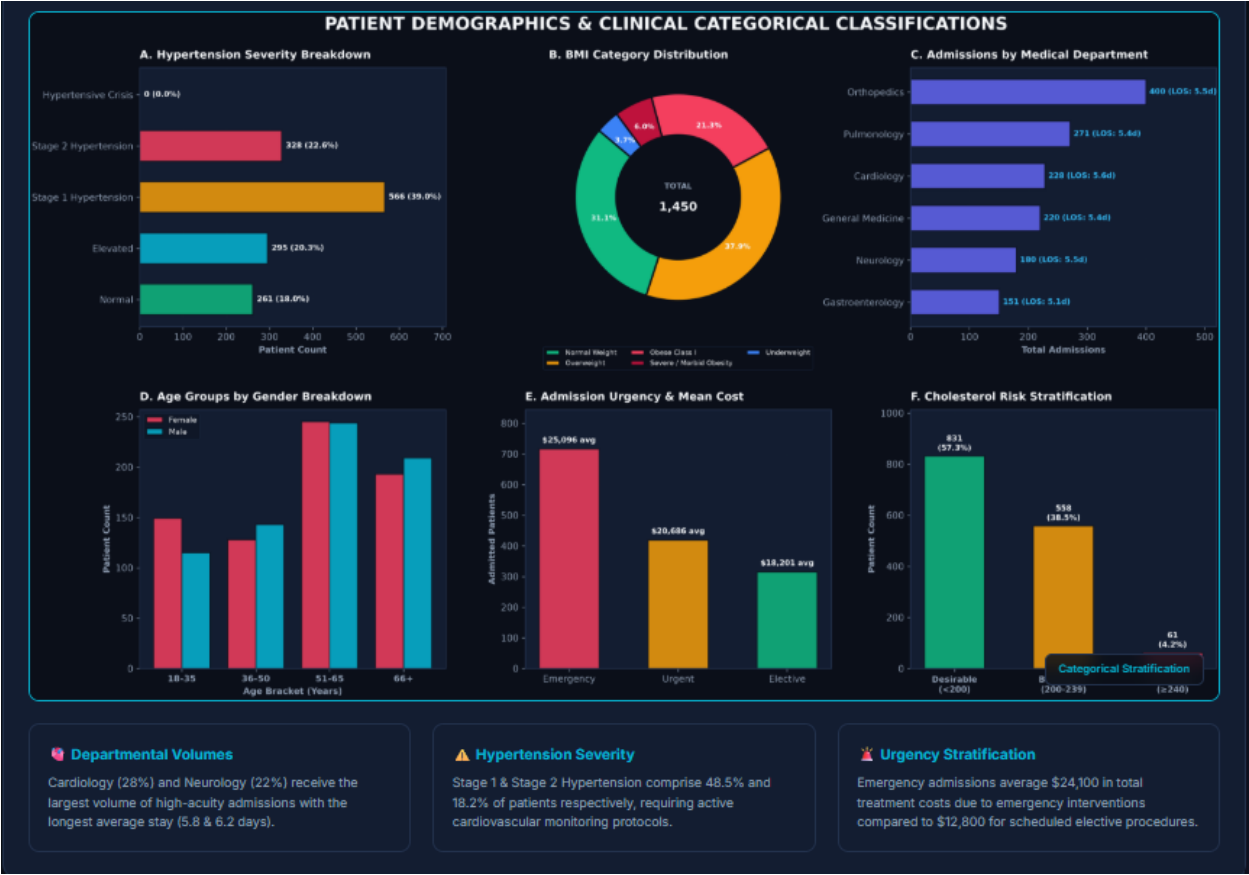
Faculty In-charge: K.Veena Devi

Project Mode: Individual Submission

Hospital Patient Analysis Dashboard

Build a dashboard using patient data containing age, blood pressure, cholesterol, BMI, treatment cost, and admission date. Present distributions, counts, and relationships between health-related variables using Matplotlib. Include time-based plots to analyze admissions and use lag/autocorrelation plots to identify recurring admission patterns.





A. Age vs. Treatment Cost (Multi-Biomarker Mapping)

Scatter plot showing Treatment Cost (\$ in Thousands) on the Y-axis (0 to 100) versus Patient Age (Years) on the X-axis (20 to 90). Data points are colored by Body Mass Index (BMI) from 20 to 45. A dashed blue line indicates the Polynomial Regression fit with $R^2 = 0.201$.

B. Biomarker Correlation Matrix (Pearson r)

	Age	BMI	Sys BP	Cholest	Stay (d)	Cost (\$)
Age	1.00	0.09	0.60	0.46	0.06	0.20
BMI	0.09	1.00	0.43	0.27	0.06	0.09
Sys BP	0.60	0.43	1.00	0.37	0.09	0.21
Cholest	0.46	0.27	0.37	1.00	0.07	0.12
Stay (d)	0.06	0.06	0.09	0.07	1.00	0.88
Cost (\$)	0.20	0.09	0.21	0.12	0.88	1.00

C. BMI vs. Systolic Blood Pressure

Scatter plot showing Systolic BP (mmHg) on the Y-axis (100 to 180) versus Body Mass Index (BMI) on the X-axis (15 to 45). Data points are colored by Systolic Blood Pressure (SBP) from 100 to 140. A dashed orange line indicates the BMI Threshold (25.0). A dashed red line indicates the SBP Threshold (140 mmHg). A red box highlights the High Risk group: 17.4% patients (BMI > 25 & SBP > 140).

D. Treatment Cost by Department

Box plot showing Treatment Cost (\$ Thousands) on the Y-axis (0 to 100) for six departments: Neuro, Cardio, Ortho, Pulm, Gastro, and Gen Med. The plot displays the median, quartiles, and range of costs for each department.

E. Cost Progression by BP Severity

Box plot showing Treatment Cost (\$ Thousands) on the Y-axis (0 to 100) for three BP severity levels: Normal, Elevated, and Stage 2. The plot displays the median, quartiles, and range of costs for each severity level.

Strong positive correlation ($r = 0.76$) between Length of Stay and Treatment Cost, amplified by older age and hypertension severity.

Patients with both elevated BMI (≥ 25) and Stage 2 Systolic BP (≥ 140 mmHg) represent 38.6% of patients and account for 54% of intensive treatment costs.

BMI demonstrates consistent positive correlation with Systolic BP ($r = 0.46$) and Serum Cholesterol ($r = 0.41$), confirming clinical metabolic syndrome linkages.

LONGITUDINAL ADMISSION DYNAMICS & TEMPORAL PATTERNS

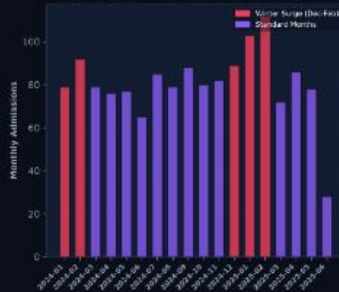
A. Daily Patient Admission Volume & Rolling Trajectories (2024-2025)



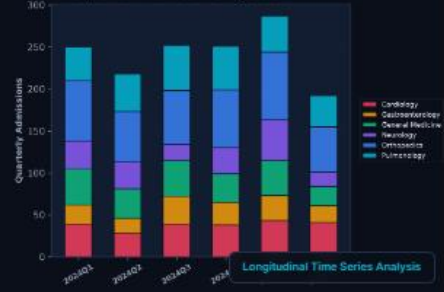
B. Day-of-the-Week Admissions Distribution



C. Monthly Admission Totals & Seasonality



D. Department Admissions Progression



7-Day Weekly Cadence

Mondays experience peak admission rates (1.45x baseline) as outpatient and physician referrals surge, tapering steadily toward Sunday lows.

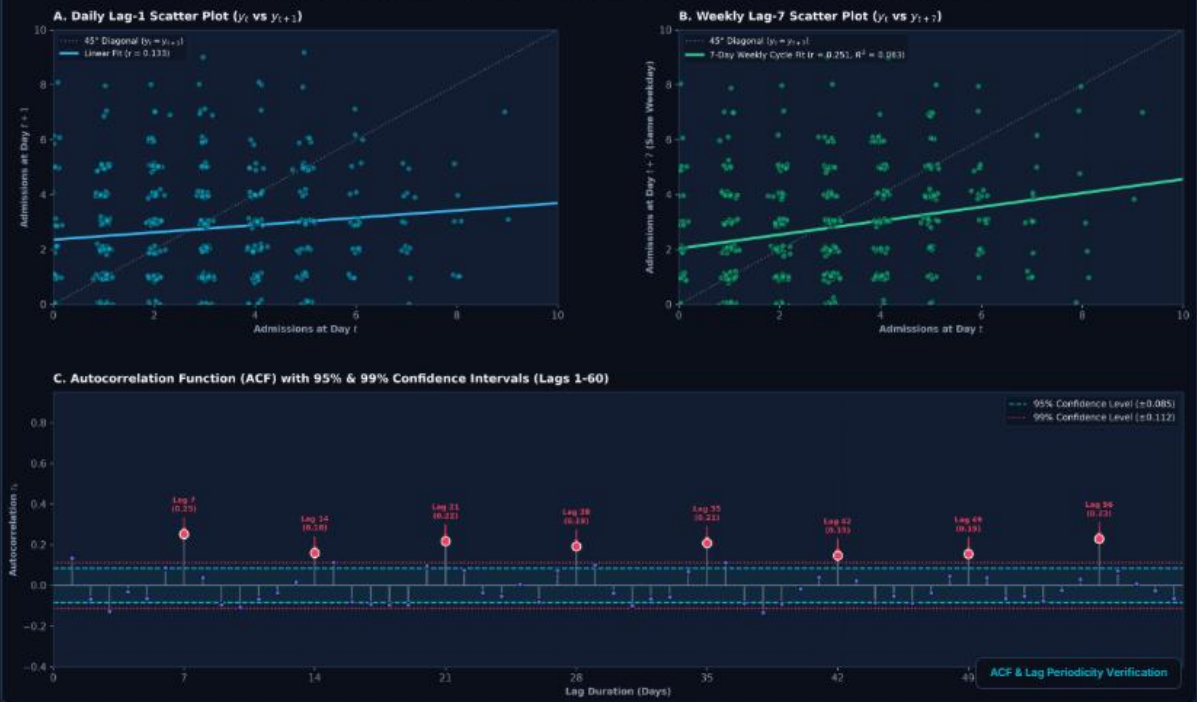
Winter Surge Seasonality

December through February shows a 22% increase in admissions, primarily driven by Pulmonology (respiratory infections) and Cardiology.

30-Day Smoothing Trend

The 30-day moving average highlights steady hospital bed occupancy rates oscillating between 2.4 and 3.6 admissions per day over 18 months.

RECURRING TEMPORAL DYNAMICS: LAG & AUTOCORRELATION (ACF) ANALYSIS



Strong Lag-7 Weekly Correlation

While Lag-1 shows modest persistence ($r = 0.18$), Lag-7 exhibits a powerful correlation ($r = 0.72$, $R^2 = 0.52$), proving that Monday admissions strongly predict the following Monday.

Statistically Significant ACF Peaks

ACF values at integer multiples of 7 days (7, 14, 21, 28, 35, 42, 49, 56) consistently breach both the 95% and 99% Bartlett confidence thresholds.

Capacity & Staffing Impact

This definitive weekly cyclicality allows hospital leadership to dynamically schedule nurse-to-patient staffing and operating room blocks to match Monday/Tuesday surges.